Resilient urgent orchestration for SAGE

An explainable scheduling policy for SAGE / Waggle nodes

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In urgent computing, the right result delivered too late is the wrong result.

THIS PROJECT

Ready applications including Mortimus → **SAGE / Waggle** adapter → **Policy engine** select · defer · reject → **NodeScheduler** creates the Pods → **Kubernetes** places and runs

SAGE remains the orchestrator — only the selection policy changes. Beehive, WES, the NodeScheduler and Kubernetes are untouched.

The problem

SAGE nodes run several applications on limited edge hardware. Routine work and time-sensitive events compete for the same CPU, memory, GPU and execution slots, while priorities change.

Three questions must be answered:

01 Which ready application should start next?

02 Can it start without exceeding safe limits?

03 Why was each application selected, deferred or rejected?

Mortimus — cameras watching a landscape for smoke — is the integration target, not a dependency.

How it decides

01 Observe and validate ready and active tasks.

02 Rank valid tasks by priority, deadline slack, queue age and success probability.

03 Admit within concurrency limits, and resource limits when capacity is trustworthy.

04 Explain every selection, deferral or rejection with a stable reason.

05 Degrade to bounded queue order if it fails.

URGENCY IS DEADLINE SLACK

```
slack = deadline - now - estimated runtime
```

What it returns

Every candidate leaves with one outcome and a stable reason. One node, one free slot, two tasks:

`smoke-detector` SELECTED

priority 90 · slack −5 s · score 80.75

`reason: selected · predictedDeadlineMiss: true`

`image-sampler` DEFERRED

priority 20 · slack +540 s · score 19.65

`reason: concurrency_limit`

Deferred stays queued for the next free slot; rejected is invalid work.

Validation path: offline replay, KWOK, k3s, shadow mode, then a controlled canary.

github.com/NunesClement/sage-resilient-urgent-scheduler · Apache-2.0

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